

# Python Cheatsheet

## MATH60033A · Session 02 · Exploratory data analysis

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Everything you need to type, in one place. The first three sections are the same every week; the last one is every name from sessions 1 to 2.

### Starting Python in VS Codium

Open the course folder once — **File > Open Folder**, then pick the repository. Everything below assumes you are inside it.

Open a terminal with **View > Terminal** (Ctrl + ` on Windows and Linux, Cmd + ` on macOS). Create the environment the first time only:

```
python3 -m venv .venv      # macOS
python3 -m venv .venv      # Linux
py -m venv .venv           # Windows (PowerShell)
```

Then activate it — **every time you open a new terminal**:

```
source .venv/bin/activate  # macOS
source .venv/bin/activate  # Linux
.venv\Scripts\activate     # Windows (PowerShell)
```

You will see (.venv) at the start of the prompt. If you do not, nothing below will find pandas. Install the packages once, with the environment active:

```
pip install -r requirements.txt
```

Finally tell the editor which Python to use: Ctrl/Cmd + Shift + P, type **Python: Select Interpreter**, and choose the one inside .venv.

### A new file, and running it

**File > New File**, then save it with a .py ending — session02.py, say. Write your code, save it, and in the terminal:

```
python session02.py      # macOS, Linux and Windows alike, inside .venv
```

Inside the environment the command is **python** on all three platforms. Outside it, macOS and Linux need **python3** and Windows needs **py**, which is one more reason to activate first.

For trying one line at a time, type **python** on its own to get the >>> prompt, and **exit()** to leave. The editor's **Run** triangle does the same thing as the command above.

### The four things you always do

```
import pandas as pd
```

```
# 1 - build a DataFrame
df = pd.DataFrame({"country": ["FR", "DE", "IT"],
                  "gdp_pc_eur": [37_000, 46_000, 32_000]})
```

```
# 2 - look at what is in it
df.head()      # the first five rows
df.shape       # (rows, columns)
df.dtypes      # what type each column holds
```

```

df.info()          # types, and how many values are missing
df.describe()      # count, mean, sd and quartiles, per numeric column

# 3 - read a file in
df = pd.read_csv("data.csv")
df = pd.read_excel("data.xlsx")          # needs openpyxl
df = pd.read_parquet("data/spine/core.parquet")

import qmib
df = qmib.load("core")                   # a course dataset, by name

# 4 - write a CSV out
df.to_csv("out.csv", index=False)      # index=False, or you get a stray column

```

### What you have learned so far

Sessions 1 to 2, in the order you met them. Every line below has run on the course data.

## Session 1 · Syllabus and Europe 2031

The workstation itself — the editor, the environment, and loading a course dataset with one line. That is section 1 above.

## Session 2 · Exploratory data analysis

```

# -- Load and shape -----
core = qmib.load("core")          # a course dataset, by name
core = pd.read_parquet("data/spine/core.parquet")
d = core.dropna(subset=["gdp_pc_eur", "productivity_idx"])
# subset= matters: without it a row missing ANY column goes
both = left.merge(right, on=["geo", "time"], how="inner")
n_rows, n_cols = core.shape

# -- Describe -----
gdp.mean(), gdp.median()          # the gap between them is the finding
gdp.std(ddof=1), gdp.var(ddof=1)  # ddof=1 divides by n-1
gdp.quantile([0.25, 0.5, 0.75])  # IQR is the third minus the first
from scipy import stats
stats.trim_mean(gdp, 0.05)        # mean after cutting 5% off each tail

```